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Supplemental information

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Aashish Satyajith and Mukund Thattai

Supporting Material

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Aashish Satyajith[1,*] and Mukund Thattai[2]

[1]Chennai Mathematical Institute, Chennai, India
[2]Simons Centre for the Study of Living Machines, National Centre for Biological Sciences, TIFR, Bangalore, India
[*]Correspondence: aashish@cmi.ac.in

APPENDIX

A. Finding optimal splits and residence times for single glycan manufacture

We used `differential_evolution()` function from the SciPy library (1) with the parameters `bounds=[(1e-7, 100), (1e-7, 100)], maxiter=200, tol=1e-10, seed=42, polish=True` to find the optimal residence times for a given split and under a given total residence time constraint.

The yield objective is a log-concave function of residence times for non-overlapping splits

A positive function $f : \mathbb{R} \to \mathbb{R}_+$ is log-concave if the logarithm of the function is concave, i.e. $\log f(\theta x + (1-\theta)y) \geq \theta \log f(x) + (1-\theta) \log f(y)$ for $0 \leq \theta \leq 1; x, y \in \mathbb{R}$ (2). A differentiable function f is concave in an interval if its first derivative is monotonically decreasing in that interval. From the definition of log-concavity, it follows that if f can be decomposed into a product of log-concave functions, then f is log-concave.

A function $f(x_1, \ldots, x_N)$ is said to be pointwise log-concave if for all fixed $x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_N$, the function is log-concave in $x_i, \forall i$. We prove that the yield function $yield(\mathcal{S}, \mathcal{T})$ for a glycan with fixed non-overlapping split $\mathcal{S}$ is point-wise log-concave. Since the yield function comprises products of yields corresponding to splits in each compartment, it suffices to prove that the expression for yield in a single compartment is a log-concave function of the residence time for that compartment. We do this by induction.

Base case: The yield corresponding to a single reaction is $(1 - e^{-t}) \cdot e^{-qt}$ for some q (t is the residence time). Both the terms can be shown to be log-concave by taking their respective logarithms and seeing that the second derivative with respect to time is non-positive. Therefore their product is also log-concave.

Inductive case: A split can have several independent components, and the final yield is calculated as the product of yields of individual components (multiplied by the undesirable reaction term, which is log-concave). We show that yield of formation of each component is a log-concave function of the residence time.

Let $g(B_i, t)$ denote the single-compartment yield of formation of a sub-tree B_i attached to a root node under residence time t. Assume by induction that $g(B_i, t)$ is log-concave in t. The overall yield of formation of the branch containing B_i is calculated as

$$\int_0^t e^{-t_1} g(B_i, t - t_1)\, \mathrm{d}t_1$$

e^{-t} is log-concave as a function of t, and $g(B_i, t)$ is log-concave by the induction hypothesis. Then the above expression represents the convolution of two log-concave functions, which is also known to be log-concave (2).

Enumerating valid splits for a two-compartment Golgi

We argue that it suffices to consider complements of connected subtrees of the first split which include the root, for the second split. The first split has to be connected because of the assumption that glycans are grown by adding one monosaccharide at a time to the component with the root. The second split must at least include the complement of the first split (otherwise the full tree will not get made). Now suppose that the second split is not a complement of a connected subtree of the first split. Hence there is a component connected to the root (parent component) and at least one component that is not (child component). Then there is a set of edges from first split that can restore connectivity of the child component to the parent component. The edges of this set are useful in the second compartment only if the child component(s) can be made in the second compartment. Since the second split is the complement of the parent and child component(s), the child component(s) cannot be made in the second compartment. Hence the edges of the set are not useful, and the split is sub-optimal because of the undesirable reaction term.

B. Generating Pareto-optimal solutions for multiple glycan manufacture

If there are objectives $f_1(\boldsymbol{x}), \ldots, f_K(\boldsymbol{x}); f_i(\boldsymbol{x}) : \mathbb{R}^n \to \mathbb{R}$ for some n, then it is known that solutions of

$$\max_{\boldsymbol{x}} \sum_{i=1}^{K} c_i\, g(f_i(\boldsymbol{x})); \sum_{i=1}^{K} c_i = 1 \text{ and } c_i \geq 0$$

where $g(\cdot)$ is a monotonically increasing function, are Pareto-optimal. In other words, maximizers of convex combinations of monotonically increasing functions of the given objectives, are Pareto-optimal (3, 4). We used `differential_evolution()` function from the SciPy library (1) with the parameters `bounds=[(1e-7, 100), (1e-7, 100)], maxiter=200, tol=1e-10, seed=42, polish=True` to optimize any convex combination of log-yields in the manuscript.

We enumerate all possible splits for all glycans and use exhaustive search to pick the Pareto-optimal solutions. For generating candidates solutions for two-glycan manufacture (Fig. 4 of main text), we generated 401 weights uniformly between 0 and 1.

C. Generating Pareto-optimal solutions by exhaustive enumeration

There were 159, 28, and 180 possible splits for each of the trees in Fig. 2A of main text, including splits where the entire tree is built in one compartment. In case of three glycan manufacture, this means there are $159 \times 28 \times 180 = 801,360$ possible combinations of splits. To bring down the combination of splits to consider, we evaluated the yields for all the splits of the three glycans, across a grid of (t_1, t_2) values. It suffices to sample timepoints between the minimum and maximum residence times across all splits of the glycans, since yields outside these bounds will be Pareto-dominated:

Claim: Let t^i_{min} and t^i_{max} denote the minimum and maximum optimal residence time in a compartment i across all splits of K glycans to be manufactured. Then $t < t^i_{\text{min}}$ and $t > t^i_{\text{max}}$ cannot be a Pareto-optimal residence time for that compartment.

Proof: It suffices to prove the claim for a single compartment. Consider the log-yields $f_1(t^i_{\text{min}}), \ldots, f_K(t^i_{\text{min}})$ obtained in that compartment for each of the K glycans at $t = t^i_{\text{min}}$. Each of the K glycans has optimal residence time greater than or equal to t^i_{min}. Consider a $t_1 < t^i_{\text{min}}$. By the concavity of the log-yields, for the jth glycan, $f_j(t_1) \leq f_j(t^i_{\text{min}}) - f'_j(t^i_{\text{min}})(t^i_{\text{min}} - t_1)$ where $f'_j(t^i_{\text{min}})$ represents the derivative of f_j at t^i_{min} (f_j is differentiable by the differentiability of log function, and the fundamental theorem of calculus). The term $f'_j(t^i_{\text{min}})(t^i_{\text{min}} - t_1) > 0$ for all the glycans whose optima is not at t^i_{min}. Thus $f_j(t_1) \leq f_j(t^i_{\text{min}}), \forall j \in \{1, \ldots, K\}$, and there will exist one or more j for which the inequality is strict because of the positive term. Hence, the yields attained at t_1 are Pareto-dominated by those at t^i_{min}, and t_1 will not be Pareto-optimal. An analogous argument can be made for $t_1 > t^i_{\text{max}}$.

We sampled a uniformly spaced grid of 251×251 values between 0 and maximum residence time rounded to the next integer, for various q's. For $q = 0.001, 0.01$ and 0.1, the maximum residence times we considered were 5, 10, and 14 respectively. We then obtained the maximum yields for the three glycans at each (t_1, t_2) pair. We retained only those splits which produced maximum yield for at least one (t_1, t_2) pair. This left a maximum of 3, 5 and 4 splits respectively for the three trees across various q's. We also considered candidate solutions generated by pairwise convex combinations of the log-yields of retained splits (this was needed to filter out any spurious Pareto-optimal solutions outside the boundaries formed by pairwise combinations of splits). After obtaining the maximizer for a convex weighting, we computed the maximum yield attainable for the third glycan at that maximizer. We considered $10,001$ weights uniformly spaced between 0 and 1 while considering pairwise combinations of the retained splits . The $10,001$ candidates for each of the maximum $3 \times 5 \times 4 = 60$ combinations along with the candidate solutions from the grid were screened for non-dominated yields, and the Pareto set was plotted in Fig. 5 of main text and Fig. S4. The corresponding yields are shown in Figs. S5, S6, and S7.

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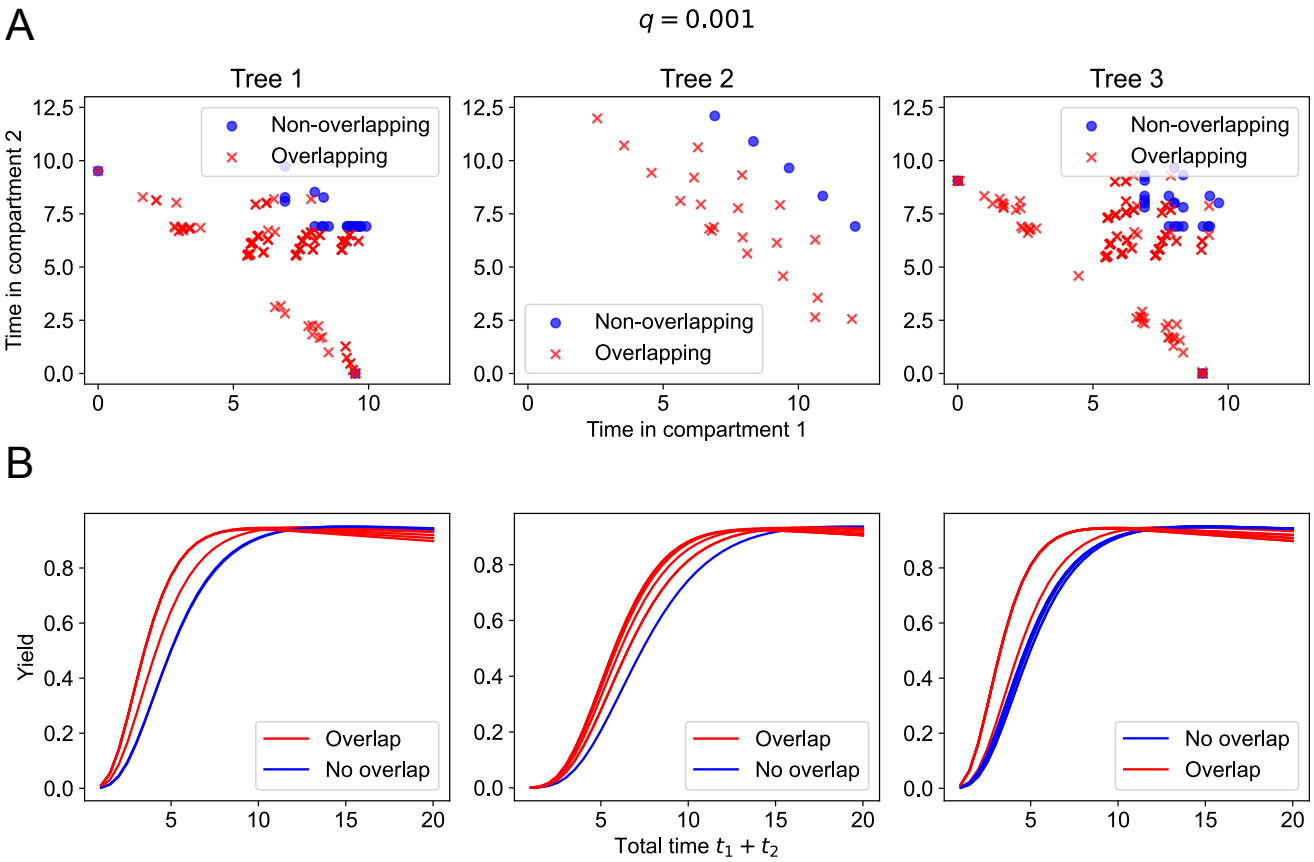


Figure S1: Optimal residence times, and optimal yields of different splits under total residence times for $q = 0.001$. (A) Optimal residence times for all splits of the three trees shown in Fig. 2C of main text. Splits with overlapping enzymes are coloured red, whereas those that do not have overlapping enzymes are coloured blue. Some splits have the same t_1 or t_2 when they have structurally equivalent growths in one of the compartments. The optimal residence times for some overlapping splits is zero in one of the compartments, even when there are enzymes in that compartment. (B) Yields of the three trees under a specified total residence time. For clarity, only splits that are optimal at some total residence times are shown. Splits which did not have any reactions in one of the compartments were removed.

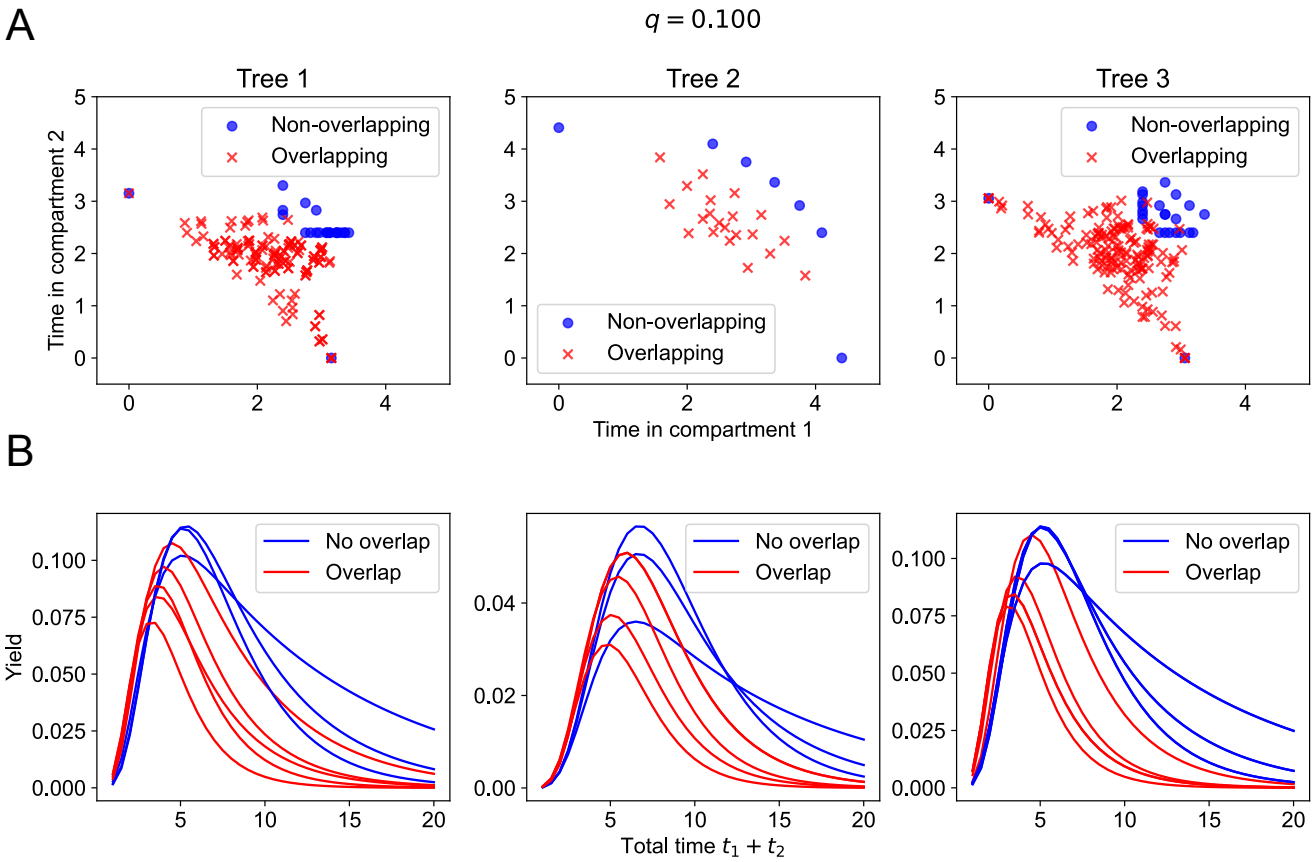


Figure S2: Optimal residence times, and optimal yields of different splits under total residence times for $q = 0.1$. (A) Optimal residence times for all splits of the three trees shown in Fig. 2C of main text. Splits with overlapping enzymes are coloured red, whereas those that do not have overlapping enzymes are coloured blue. Some splits have the same t_1 or t_2 when they have structurally equivalent growths in one of the compartments. The optimal residence times for some overlapping splits is zero in one of the compartments, even when there are enzymes in that compartment. (B) Yields of the three trees under a specified total residence time. For clarity, only splits that are optimal at some total residence times are shown. Splits which did not have any reactions in one of the compartments were removed.

Color	Tree 1 Comp. 1	Tree 1 Comp. 2	Tree 1 Overlap	Tree 2 Comp. 1	Tree 2 Comp. 2	Tree 2 Overlap
	$0 \to 1, 1 \to 3$	$0 \to 2, 1 \to 4, 3 \to 5, 3 \to 6$	No	$0 \to 1, 1 \to 2, 2 \to 3$	$3 \to 4, 4 \to 5, 5 \to 6$	No
	$0 \to 1, 1 \to 3$	$0 \to 2, 1 \to 4, 3 \to 5, 3 \to 6$	No	$0 \to 1, 1 \to 2, 2 \to 3, 3 \to 4$	$2 \to 3, 3 \to 4, 4 \to 5, 5 \to 6$	Yes
	$0 \to 1, 1 \to 3$	$0 \to 2, 1 \to 4, 3 \to 5, 3 \to 6$	No	$0 \to 1, 1 \to 2, 2 \to 3, 3 \to 4$	$3 \to 4, 4 \to 5, 5 \to 6$	Yes
	$0 \to 1, 1 \to 3$	$0 \to 2, 1 \to 4, 3 \to 5, 3 \to 6$	No	$0 \to 1, 1 \to 2, 2 \to 3, 3 \to 4$	$4 \to 5, 5 \to 6$	No
	$0 \to 1, 0 \to 2, 1 \to 3$	$1 \to 4, 3 \to 5, 3 \to 6$	No	$0 \to 1, 1 \to 2, 2 \to 3$	$3 \to 4, 4 \to 5, 5 \to 6$	No

Table S1: Pareto-optimal enzyme splits for two-glycan manufacture. Each row corresponds to a combination of splits for the two trees. Tree 1 is labeled as $\{0 \to 1, 0 \to 2, 1 \to 3, 1 \to 4, 3 \to 5, 3 \to 6\}$, and Tree 2 is labeled as $\{0 \to 1, 1 \to 2, 2 \to 3, 3 \to 4, 4 \to 5, 5 \to 6\}$. The columns labeled "Comp. 1" and "Comp. 2" specify the reactions in the first and second compartments respectively, and the column labeled "Overlap" specifies whether the splits for the two trees have any common reactions within the compartments.

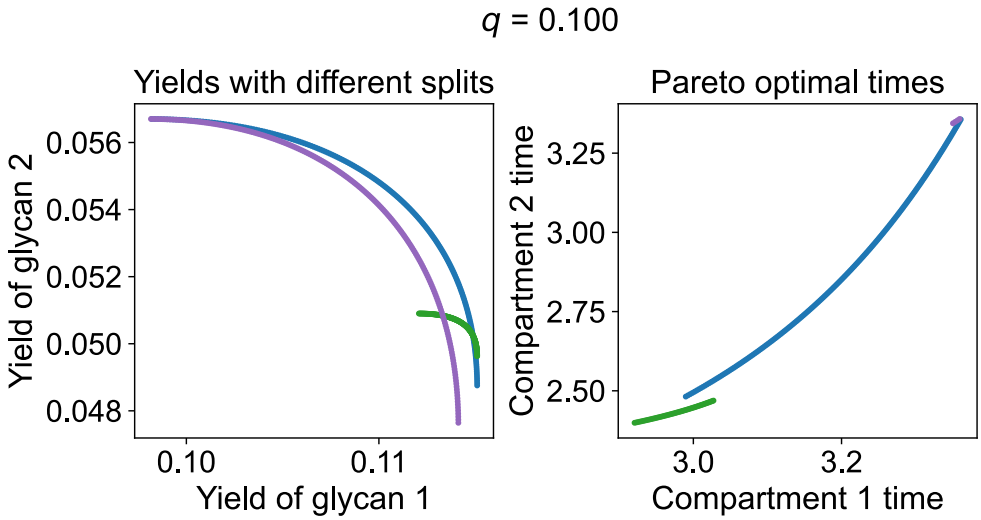


Figure S3: Pareto-optimal splits and residence times for two-glycan manufacture at $q = 0.1$. The colour of the scatter denotes the split that was used. The splits corresponding to the colours are shown in Table S1. At relatively higher values of yield of glycan 1, the optimal split switches from the split combination coloured blue to one coloured green.

Color	Comp. 1	Comp. 2	Overlap?
	$0 \to 1, 0 \to 2$	$1 \to 3, 1 \to 4, 3 \to 5, 3 \to 6$	No
	$0 \to 1, 1 \to 3$	$0 \to 2, 1 \to 4, 3 \to 5, 3 \to 6$	No
	$0 \to 1, 0 \to 2, 1 \to 3$	$1 \to 4, 3 \to 5, 3 \to 6$	No

Table S2: Pareto-optimal enzyme splits for Tree 1 in case of three-glycan manufacture, where Tree 1 is labeled as $\{0 \to 1, 0 \to 2, 1 \to 3, 1 \to 4, 3 \to 5, 3 \to 6\}$. The columns labeled "Comp. 1" and "Comp. 2" specify the reactions in the first and second compartments respectively, and the column labeled "Overlap" specifies whether the splits have any common reactions in the two compartments.

Color	Comp. 1	Comp. 2	Overlap?
	$0 \to 1, 1 \to 2, 2 \to 3$	$2 \to 3, 3 \to 4, 4 \to 5, 5 \to 6$	Yes
	$0 \to 1, 1 \to 2, 2 \to 3$	$3 \to 4, 4 \to 5, 5 \to 6$	No
	$0 \to 1, 1 \to 2, 2 \to 3, 3 \to 4$	$2 \to 3, 3 \to 4, 4 \to 5, 5 \to 6$	Yes
	$0 \to 1, 1 \to 2, 2 \to 3, 3 \to 4$	$3 \to 4, 4 \to 5, 5 \to 6$	Yes
	$0 \to 1, 1 \to 2, 2 \to 3, 3 \to 4$	$4 \to 5, 5 \to 6$	No

Table S3: Pareto-optimal enzyme splits for Tree 2 in case of three-glycan manufacture, where Tree 2 is labeled as $\{0 \to 1, 1 \to 2, 2 \to 3, 3 \to 4, 4 \to 5, 5 \to 6\}$. The columns labeled "Comp. 1" and "Comp. 2" specify the reactions in the first and second compartments respectively, and the column labeled "Overlap" specifies whether the splits have any common reactions in the two compartments.

Color	Comp. 1	Comp. 2	Overlap?
	$0 \to 1, 0 \to 2$	$0 \to 3, 1 \to 4, 2 \to 5, 4 \to 6$	No
	$0 \to 1, 0 \to 2, 0 \to 3$	$1 \to 4, 2 \to 5, 4 \to 6$	No
	$0 \to 1, 0 \to 2, 1 \to 4$	$0 \to 3, 2 \to 5, 4 \to 6$	No
	$0 \to 1, 0 \to 2, 0 \to 3, 1 \to 4$	$2 \to 5, 4 \to 6$	No

Table S4: Pareto-optimal enzyme splits for Tree 3 in case of three-glycan manufacture, where Tree 3 is labeled as $\{0 \to 1, 0 \to 2, 0 \to 3, 1 \to 4, 2 \to 5, 4 \to 6\}$. The columns labeled "Comp. 1" and "Comp. 2" specify the reactions in the first and second compartments respectively, and the column labeled "Overlap" specifies whether the splits have any common reactions in the two compartments.

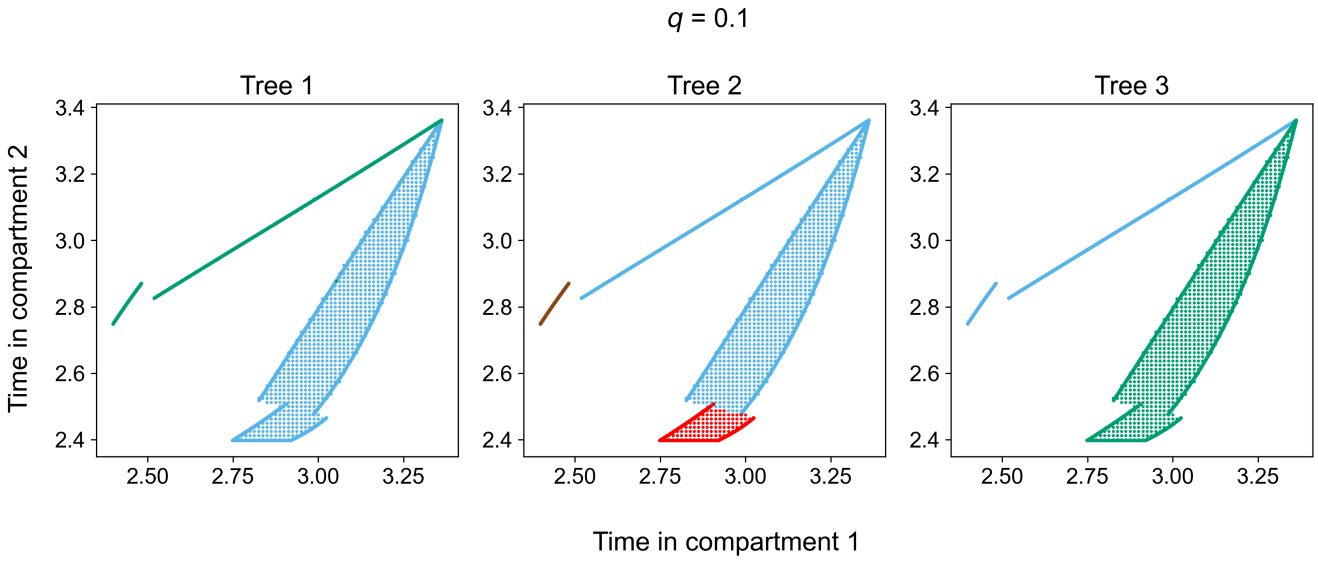


Figure S4: Pareto-optimal splits and residence times for three-glycan manufacture at $q = 0.1$. The colour of the scatter denotes the split that was used; Tables S2, S3, and S4 specify the splits corresponding to each colour for the three trees.

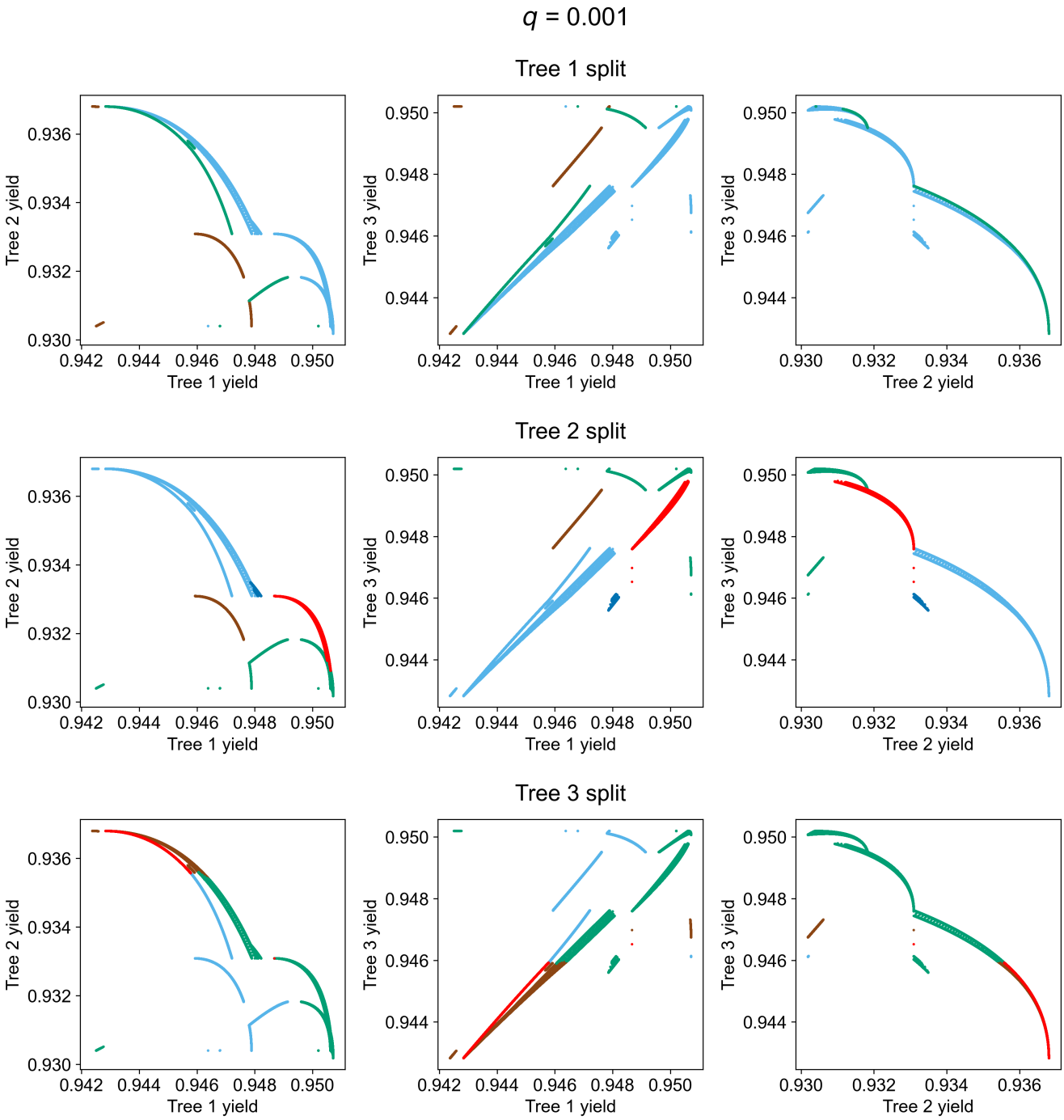


Figure S5: Pareto-optimal yields for three-glycan manufacture at $q = 0.001$. The colour of the scatter denotes the split that was used; Tables S2, S3, and S4 specify the splits corresponding to each colour for the three trees.

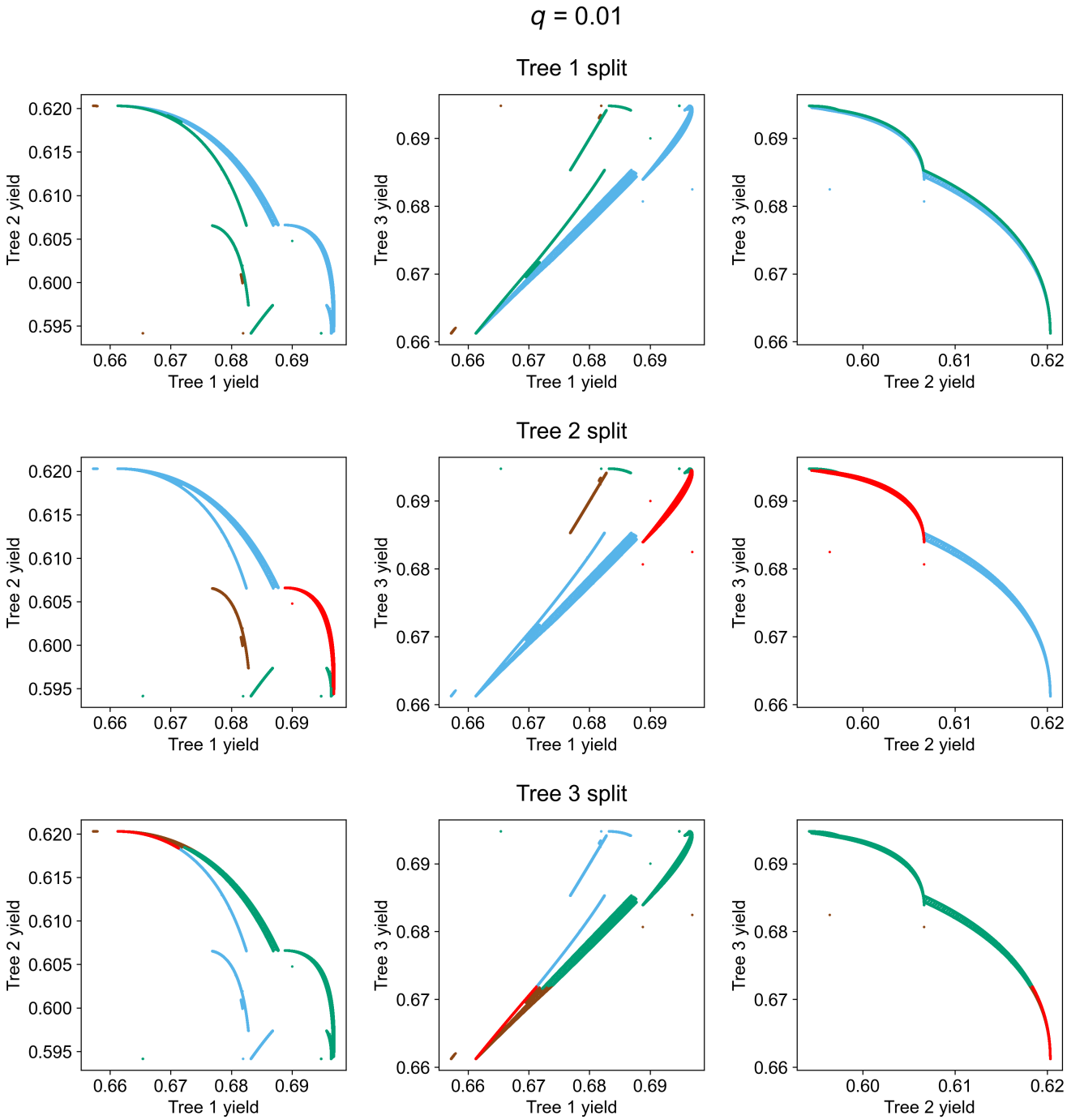


Figure S6: Pareto-optimal yields for three-glycan manufacture at $q = 0.01$. The colour of the scatter denotes the split that was used; Tables S2, S3, and S4 specify the splits corresponding to each colour for the three trees.

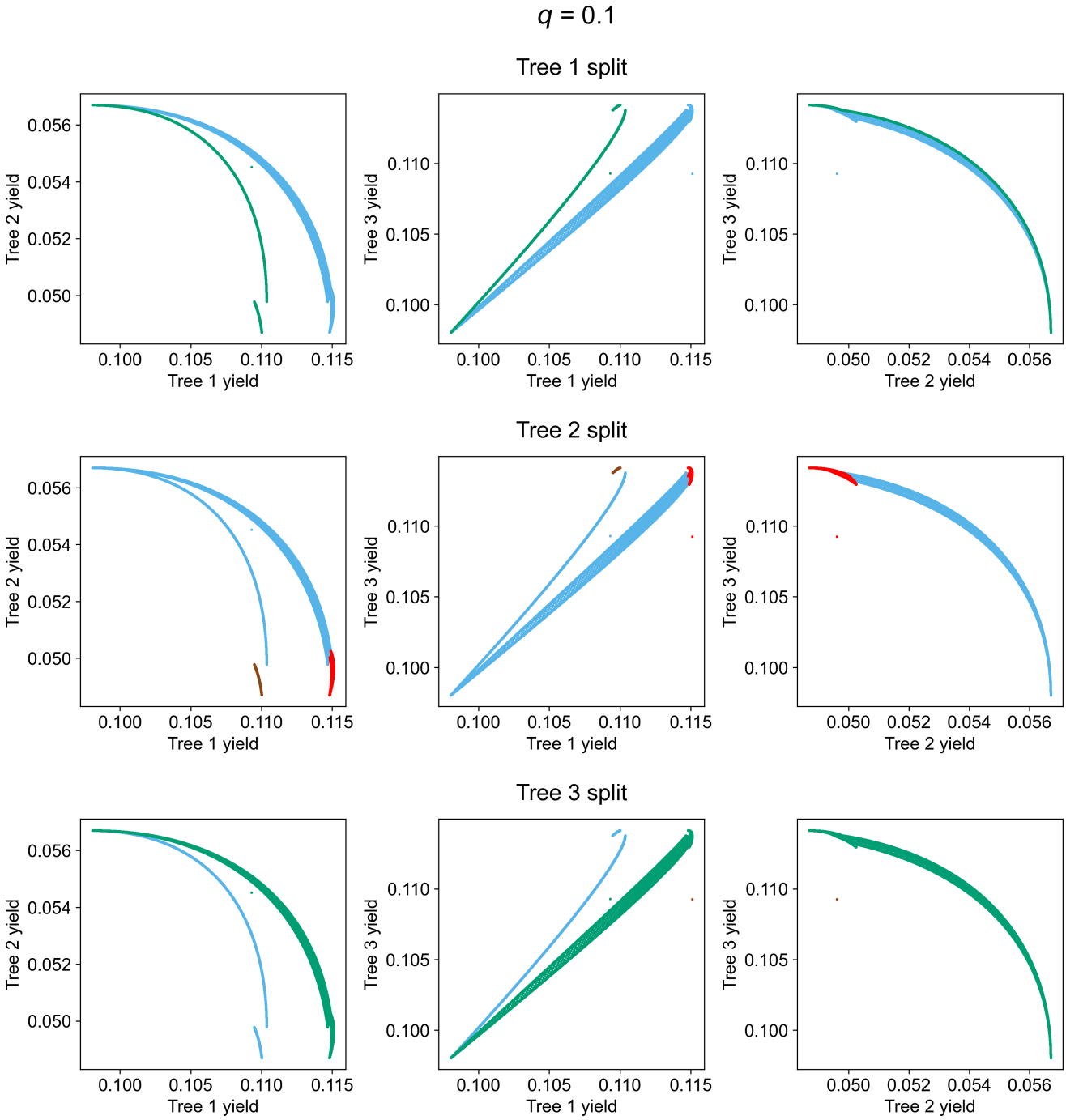


Figure S7: Pareto-optimal yields for three-glycan manufacture at $q = 0.1$. The colour of the scatter denotes the split that was used; Tables S2, S3, and S4 specify the splits corresponding to each colour for the three trees.